

Element F

Requirement	Viability	Evidence	Risks
Safety	The design meets this requirement. The prototype is made with a sticky adhesive that ensures the lights will not fall off the ceiling/wall.	Many LED lights that can be bought online are also made with sticky adhesive and most likely last a long time.	There is a possibility that the quality of the adhesive on the lights we order are not sticky enough, so to replace that we could use some sort of sticky tack or tape to keep them on the wall. We could also design a clip to attach the lights.
Aesthetic/creates diverse mood	The design meets this requirement. The lights use a variety of different colors that make the room look aesthetic.	There are many LED light strips that people can buy that include many different colors that can be changed based on the person's mood.	The lights might not have all the colors that the user wants, but this can be solved if we can program them to add more colors.
Durability	The design has yet to meet this requirement because we do not know how long the lights are going to last.	Previous reviews of LED lights strips say that they last around 4-6 years.	The lights might not last as long as anticipated. The LEDs may wear out, lose their intensity, or the tape could wear as well.
Entertaining and effectiveness	The design meets this requirement. The lights have different modes where they can do different patterns.	Based on research, LED lights can enhance the mood and experience at events or when someone is alone.	There is a chance that people who use LED lights get bored of them easily, which is why we are programming fun patterns to the lights.
Adaptability to different needs	The lights meet this requirement. The remote and codes allow for them to be easily changed to the users personal needs.	LED light strips have a variety of color options that people can change to what they like.	Some people might have very specific needs that might not be met with the lights that we make.
Ease of Use	The design meets this requirement. It uses a remote controller that can easily be used to	The remote that corresponds with different types of LED light strips are very	It's possible that the remote we decide to use is confusing for some people, but the goal is to

	switch the colors of the lights.	simple and easy to understand.	make it as simple as possible.
Number of Zones	The design meets this requirement. The arduino allows for coding of different colors in different areas of the strip	We tested this on a past design using the same LED lights. The LED lights were placed in a grid and we were able to zone the columns.	The amount of zones we design is based on the stakeholders needs, so if we don't know we may make too many or too little.
Affordability	The design meets this requirement. Lights are below our limit and we did research on which LED lights would be affordable and effective.	There are multiple options of light strips that meet or are below our budget.	There might be better quality lights that are above our budget, but we made sure that the lights we ordered meet our budget and are good.